

(a) Design a triangular wave generator with 80% Duty cycle

$$\text{We know, Duty cycle} = \frac{T_1}{T_1 + T_2}$$

$$\begin{aligned} &= \frac{\frac{V_{TH} - V_{TL}}{V_H}}{\frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L}} \\ &= \frac{1/V_H}{1/V_H - 1/V_L} \end{aligned}$$

$$\therefore 0.8 = \frac{1}{1 - \frac{V_H}{V_L}}$$

$$\Rightarrow \frac{V_H}{V_L} = -\frac{2}{8}$$

Therefore, let $V_H = 2V$, then $V_L = -8V$

All other component values can be assumed freely.

(b) Implement the following function using CMOS Logic

$$Y = \overline{A}B + \overline{C}\overline{D}$$

Simplifying, $Y = \overline{A}B + \overline{C} + D$

